Programming in C: Class Test

Decision Making and Iterative Statements

Name: _____	Roll No: _____
Class / Section: _____	Date: _____
Time: 90 Minutes	Maximum Marks: 40

Instructions:

- Attempt **all four** questions. Each question carries 10 marks.
- Write your programs in the **C language**. Read input exactly as described and print output exactly in the given format.
- Arrays are not required for any question.
- Marks are awarded for correct logic, correct output format and handling of boundary cases.

Question 1: Parcel Shipping Cost [10 Marks]

Topic: switch statement, conditional operator

Problem Statement:

A courier company charges for parcels by delivery zone. The zone is given as a code: **L** (Local) costs Rs. 20 per kg, **N** (National) costs Rs. 45 per kg, and **I** (International) costs Rs. 120 per kg. If the parcel weighs more than 5 kg, a flat handling fee of Rs. 50 is added to the total. For any other zone code, the program must print **INVALID ZONE**. Write a program that reads the zone code and the weight, then prints the shipping cost.

Input Format:

- A character **zone** (L, N or I), followed by a float **weight** in kg, on one line.

Output Format:

- **Shipping Cost: X**, where X has 2 decimal places, or **INVALID ZONE**.

Constraints:

- 0 < weight <= 100
- zone is an uppercase letter
- Cost = rate x weight, plus Rs. 50 if weight > 5

Sample Input

```
N 4.5
```

Sample Output

```
Shipping Cost: 202.50
```

Explanation:

- Zone N has a rate of Rs. 45 per kg, so the cost is 45 x 4.5 = 202.50.
- The weight is not above 5 kg, so no handling fee is added.

Question 2: Tea Stall Vending Machine [10 Marks]

Topic: Menu-driven programming, do-while loop, switch statement

Problem Statement:

A vending machine at a tea stall shows this menu repeatedly until the customer chooses Checkout:
1. Tea (Rs. 10 each) 2. Coffee (Rs. 15 each) 3. Milk (Rs. 12 each) 4. Checkout
For choices 1, 2 and 3, the customer then enters the quantity. For any other choice, print **INVALID CHOICE**. At checkout, if the total is Rs. 100 or more, a 10% discount is applied. Write a menu-driven program that processes the choices and prints the final bill.

Input Format:

- A sequence of integers. Each choice is on its own line. For choices 1 to 3, the next line is the quantity. The input ends after choice 4.

Output Format:

- Choice 1: **Added Tea xQ**
- Choice 2: **Added Coffee xQ**
- Choice 3: **Added Milk xQ**
- Invalid choice: **INVALID CHOICE**
- Choice 4: **TOTAL: T** (2 decimals), followed by **DISCOUNT APPLIED** if the discount was given.

Constraints:

- 1 <= quantity <= 20
- At most 50 menu operations before checkout.

Sample Input

```
1
2
2
3
4
```

Sample Output

```
Added Tea x2
Added Coffee x3
TOTAL: 65.00
```

Explanation:

- Two teas cost 20 and three coffees cost 45, so the total is 65.
- The total is below 100, so no discount is applied.

Question 3: Exam Result Statistics [10 Marks]

Topic: for loop, if-else

Problem Statement:

After an exam, a teacher enters the marks of n students one by one. A student passes with 40 marks or more. Write a program that reads all the marks in a loop and prints the number of students who passed, the number who failed, the highest mark, and the class average. You do not need to store the marks.

Input Format:

- Line 1: an integer **n**, the number of students.
- Next n lines: an integer **marks**.

Output Format:

- One line: **Passed: p Failed: f Highest: h Average: a**, where the average has 2 decimal places.

Constraints:

- 1 <= n <= 100
- 0 <= marks <= 100
- Pass mark: marks >= 40

Sample Input

```
5
85
40
39
92
60
```

Sample Output

```
Passed: 4 Failed: 1 Highest: 92 Average: 63.20
```

Explanation:

- Marks 85, 40, 92 and 60 are 40 or above, so 4 students passed. Only 39 is a fail.
- The highest mark is 92. The sum is 316 and 316 / 5 = 63.20.

Question 4: Loan Eligibility Checker [10 Marks]

Topic: Nested if, else-if ladder

Problem Statement:

A bank screens personal loan applications in steps. If the applicant's age is below 21 or above 60, the application is rejected for age. Otherwise, if the monthly income is below Rs. 25,000, it is rejected for income. Otherwise the credit score decides:
• Score 750 or above: approved for 10 lakh.
• Score from 650 to 749: if income is Rs. 50,000 or more, approved for 5 lakh; otherwise sent for manual review.
• Score below 650: rejected for credit.
Write a program that reads the age, monthly income and credit score, then prints the decision.

Input Format:

- Three space-separated integers: **age**, **income**, **score**.

Output Format:

- One of: **REJECTED: AGE**, **REJECTED: INCOME**, **REJECTED: CREDIT**, **APPROVED: 10 LAKH**, **APPROVED: 5 LAKH**, **MANUAL REVIEW**.

Constraints:

- 18 <= age <= 70
- 0 <= income <= 500000
- 300 <= score <= 900

Sample Input

```
30 60000 780
```

Sample Output

```
APPROVED: 10 LAKH
```

Explanation:

- Age 30 is within 21 to 60.
- Income 60000 is at least 25000.
- Score 780 is at least 750, so the loan is approved for 10 lakh.

*** *End of Question Paper* ***